Graphs and Explanations

Here are the explanations of the metric + graph output. The answer to the questions needed in the project pdf are in the Colab file after each task. Please check there

Cell 1

Dataset Overview

The dataset contains **119,390 records** with **32 features**, representing hotel booking information. Each row corresponds to a single booking instance.

The features include a mix of:

- **Numerical variables**: e.g., `lead_time`, `adr`, `stays_in_week_nights`, `total_of_special_requests`
- **Categorical variables**: e.g., `hotel`, `meal`, `market_segment`, `deposit_type`
- **Date-related variables**: e.g., `arrival_date_year`, `arrival_date_month`
- **Target variable**: `is_canceled` (binary classification)

Initial Observations

- The dataset includes both **Resort Hotel** and **City Hotel** bookings.
- Some columns (e.g., `agent`, `company`) contain **missing values**, indicating incomplete booking metadata.
- The `adr` (Average Daily Rate) varies significantly, suggesting probable **outliers or anomalies**.
- The presence of `is_canceled` makes this a **supervised classification problem (this means it can be converted into a binary flag using labels)**.

This dataset is suitable for predicting whether a booking will be canceled based on:

- booking behavior (lead time, changes)
- customer characteristics (guests, requests)
- temporal patterns (seasonality, month)

A preview of the dataset informs us of its structure and highlights the need for cleaning and feature engineering before model training.

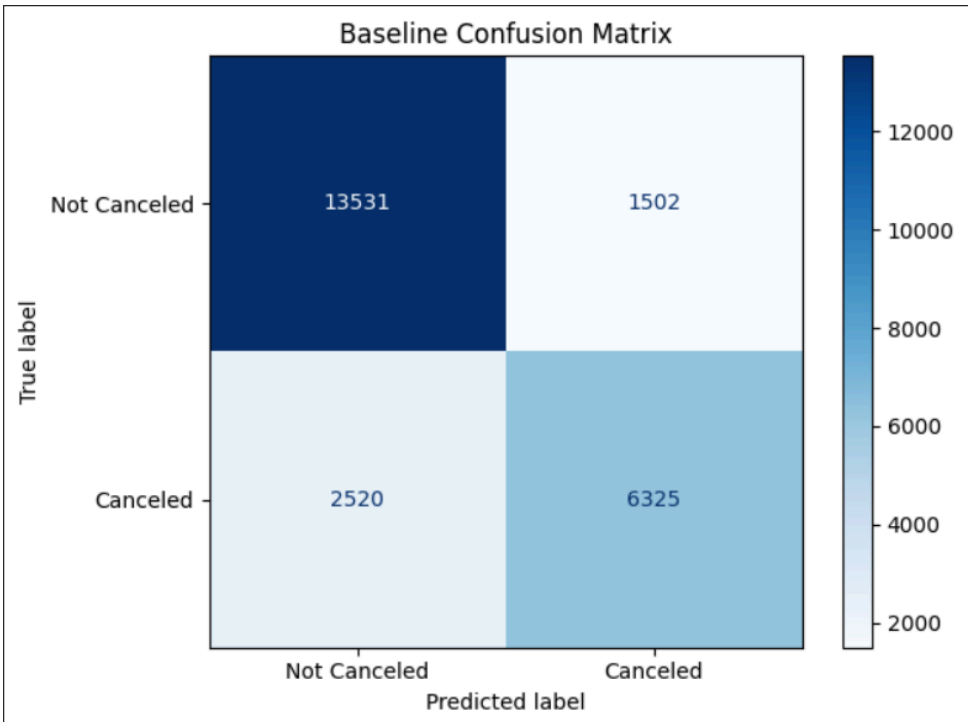
Cell 2

TASK 1

The baseline model was trained using a minimal set of features with basic preprocessing (imputation + standard scaling).

Performance Metrics

- **Accuracy**: 0.8316
- **ROC-AUC**: 0.8917
- **F1 Score**: 0.7588



Confusion Matrix Analysis

	Predicted Not Canceled	Predicted Canceled
Actual Not Canceled	13531 (TN)	1502 (FP)
Actual Canceled	2520 (FN)	6325 (TP)

Observations

- The model performs **b** even with minimal features.
- A high **ROC-AUC (0.89)** indicates good ability to distinguish between canceled and non-canceled bookings.
- However, the **F1 score (0.76)** suggests room for improvement in balancing precision and recall.

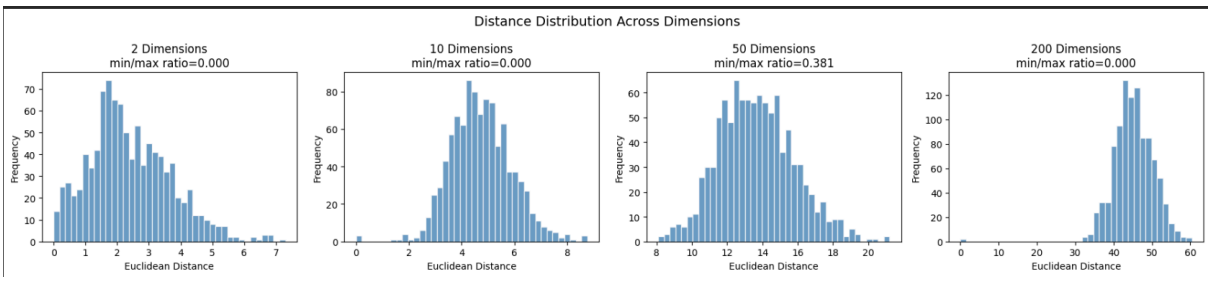
Error Analysis

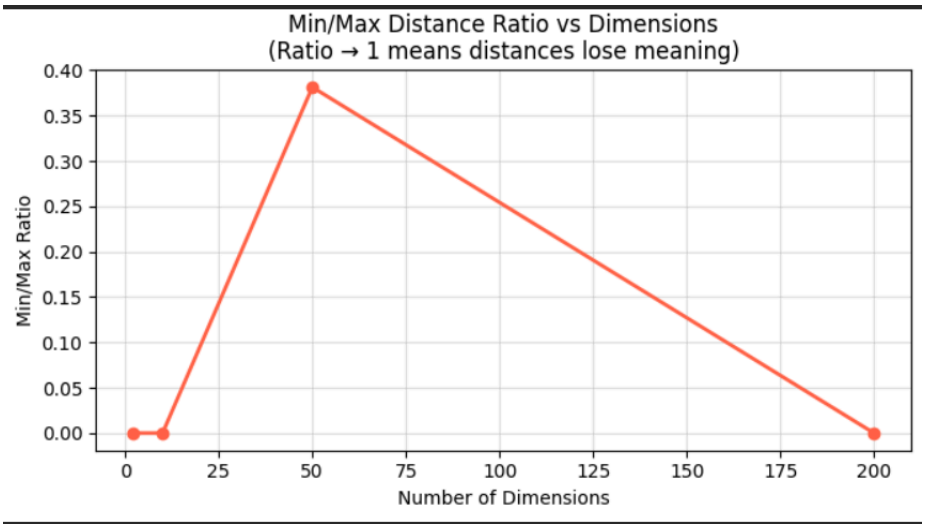
- **False Positives (1502)**: Some bookings are incorrectly predicted as canceled → could lead to unnecessary interventions.
- **False Negatives (2520)**: A larger concern — actual cancellations are missed → potential business loss.
- The model is slightly biased toward predicting **non-cancellations more confidently (it biases towards decreasing false positives)**.

It should prioritise decreasing false negatives to benefit the business model.

<u>Cell 3</u>

TASK 2





Curse of Dimensionality Analysis

To understand how distance metrics behave in higher dimensions, we analyzed the distribution of Euclidean distances across different feature dimensions (2, 10, 50, 200).

Observations from Distance Distributions

- As dimensionality increases:
  - The **mean distance increases significantly** (2.39 → 45.14)
  - The **spread (std deviation)** increases but not proportionally
  - Distances begin to **cluster closer together**
- At **low dimensions (2, 10)**:
  - Distances are more **spread out**
  - Clear distinction between near and far points
- At **higher dimensions (50, 200)**:
  - Distances become more **concentrated**
  - Histograms look tighter and more similar

Min/Max Distance Ratio Insight

Dimensions	Min/Max Ratio
2	0.0000
10	0.0000
50	0.3814
200	0.0000

- The **min/max ratio increases initially (at 50D)** → distances becoming similar
- At very high dimensions (200D), instability/noise causes irregular behavior

When the ratio approaches **1**, distances lose meaning because all points are almost equally far.

Concept

This demonstrates the **curse of dimensionality**:

- In high-dimensional space:
    - The concept of “nearest” and “farthest” becomes less meaningful
    - Distance-based models (like KNN, clustering) become less effective

Why This Matters for Our Model

- Our dataset initially has many features → risk of high dimensionality
- This justifies:
    - **Feature engineering** (to create meaningful features instead of many raw ones)
    - **Feature selection** (to remove irrelevant/redundant features)

Cell 4

Task 3: Feature Transformation & Scaling

Part A: Numerical Feature Overview

The working dataset consists of **119,389 rows and 8 numerical features**.

Key Observations

- **lead_time**
  - Mean = 104, Max = 737 → **heavily right-skewed**
  - Large std (106) → high variability
- **adr (price)**
  - Mean ≈ 101, Max = 5400 (huge)
  - Clear presence of **extreme outliers**
- **stays_in_week_nights / weekend**
  - Mostly small values → typical hotel stays are short
- **adults**
  - Median = 2 → most bookings are for couples
- **previous_cancellations & booking_changes**
  - Mostly 0 → **sparse features**, but potentially very informative

Insight

> Several features show **skewness and outliers**, indicating the need for transformation and robust scaling.

Part B: Binning / Discretization

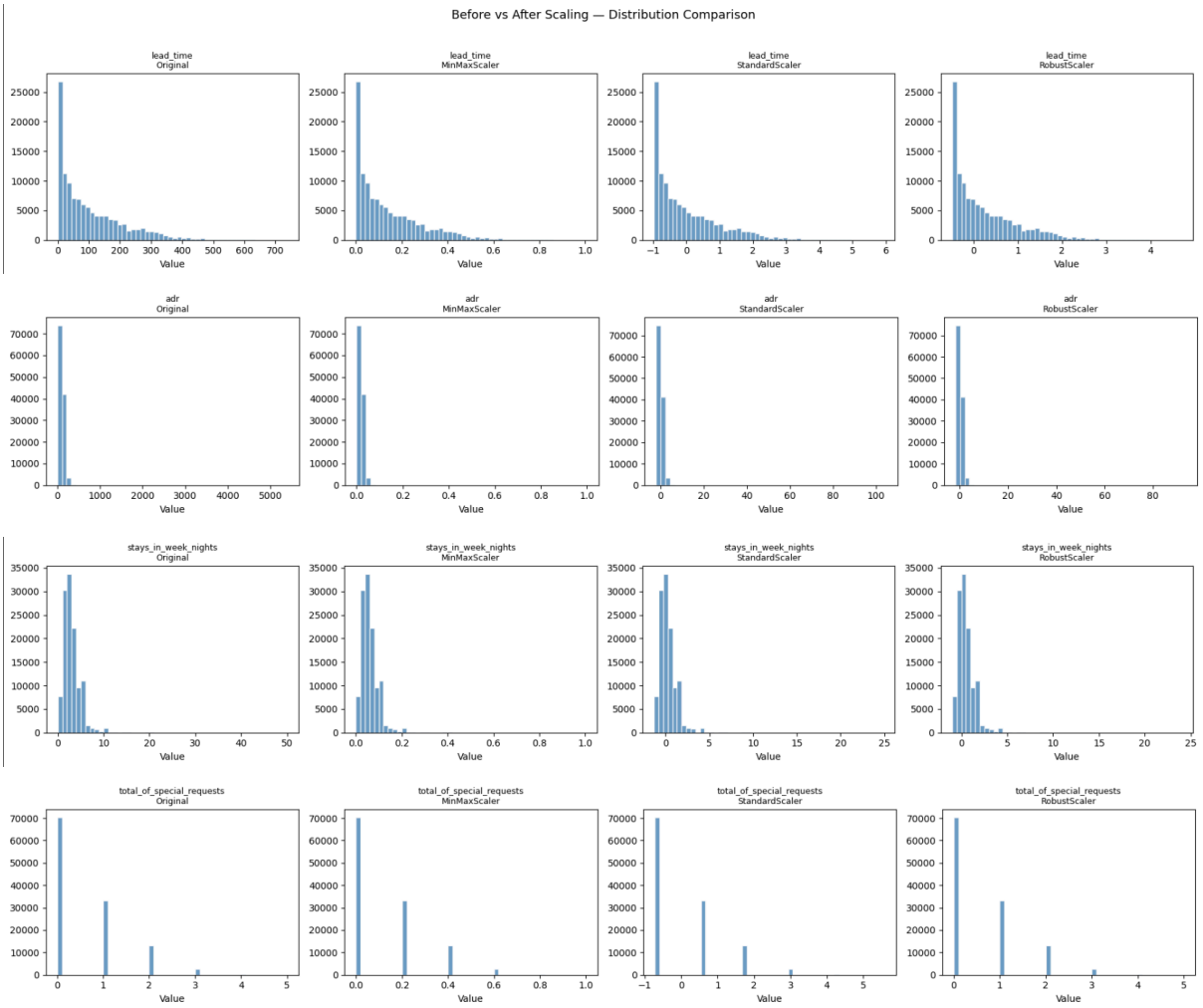
Observations

- **lead_time_bin**
  - Most bookings happen **close to arrival**
  - Very long-term bookings are rare → niche behavior
- **adr_quartile**
  - Almost perfectly balanced → ideal for modeling
  - Good feature for capturing **pricing segments**

Insight

> Binning converts continuous variables into meaningful categories, helping models capture **non-linear patterns** and improving interpretability.

Part C: Scaling Comparison



Observations

- **MinMaxScaler**
  - Compresses everything into [0,1]
  - **Sensitive to outliers** → not ideal here
- **StandardScaler**
  - Centers data but still influenced by extreme values
- **RobustScaler**
  - Uses median & IQR → **resistant to outliers**
  - Produces more stable scaling for skewed data

Given the heavy-tailed nature of features like ADR and lead_time, robust scaling is preferred over standard normalization techniques.

Cell 5

Task 4: Impact of Scaling and Distance Metrics

To evaluate the effect of feature scaling and distance metrics, a KNN classifier was tested under different configurations.

Key Observations

1) **Scaling is essential for KNN**

- Performance improves significantly after scaling:
  - Accuracy: **0.7624** → **~0.774**
  - F1: **0.6546** → **~0.673**
- Without scaling, features with large ranges (like `adr`, `lead_time`) dominate distance calculations

2) **RobustScaler performs slightly better**

- Consistently gives the **best or near-best results**
- Confirms earlier finding:
  - Dataset contains **outliers**
  - RobustScaler handles them better than StandardScaler

3) **Distance metric impact is small**

- Euclidean vs Manhattan:
  - Very minimal difference in performance
- Suggests:
  - Feature space is relatively stable after scaling
  - Model is **not highly sensitive** to metric choice here

Conclusion

- Scaling significantly improves KNN performance
- **RobustScaler + Manhattan distance** gives the best overall results (though marginally)
- Choice of distance metric is less impactful than proper scaling

Cell 6

Task 5: Model Evaluation with Cross-Validation

To ensure model robustness and avoid overfitting, **5-fold cross-validation** was performed using ROC-AUC as the evaluation metric.

Cross-Validation Results

- **ROC-AUC Scores**:
  [0.9025, 0.9040, 0.9014, 0.9046, 0.9034]
- **Mean ROC-AUC**: **0.9032**
- **Standard Deviation**: **0.0011**

Observations

- The scores are **very consistent across folds**
- Extremely low standard deviation (**0.0011**) indicates:
  - Model performance is **stable**
  - Minimal variance due to data splits

Interpretation

> The model generalizes well across different subsets of the data, showing strong robustness and low sensitivity to sampling variations.

Test Set Performance

- **Accuracy**: 0.8394
- **ROC-AUC**: 0.9009
- **F1 Score**: 0.7701

Comparison with Cross-Validation

- Test ROC-AUC (**0.9009**) is very close to CV mean (**0.9032**)
- Indicates:
  - **No significant overfitting**
  - Model performance is reliable on unseen data

Conclusion

- Cross-validation confirms that the model is:
  - **Stable**
  - **Generalizable**
- The slight drop from CV to test performance is expected.

Cell 7

Task 6: Feature Extraction

New features were derived from existing variables to capture **temporal patterns, booking behavior, and domain-specific insights**.

Extracted Features Overview

Feature	Description
arrival_season	Season derived from arrival month (e.g., Summer, Winter)
is_peak_month	Binary flag indicating peak travel months
lead_time_bucket	Categorized booking window (Last Minute, Short, Medium, Long, Very Long)
arrival_quarter	Quarter of the year (1–4)
high_cancel_country	Binary indicator for countries with historically high cancellation rates

Observations

- **Seasonality (arrival_season, arrival_quarter)**
  - Captures travel trends → cancellations often vary by season

- **Peak indicator (is_peak_month)**
    - High-demand months may influence:
        - pricing
        - cancellation behavior
- **Lead time bucket**
    - Converts a continuous variable into **behavioral segments**
    - Example:
        - “Last Minute” vs “Very Long” bookings behave very differently
- **high_cancel_country**
    - Encodes historical cancellation behavior at a **group level**
    - Adds strong predictive signals.
    - Note: The `high_cancel_country` feature is derived using aggregate statistics and must be computed using **training data only** to avoid data leakage.

Conclusion

- Extracted features introduce:
    - **temporal awareness** (season, quarter)
    - **behavioral segmentation** (lead time buckets)
    - **group-level intelligence** (country-based patterns)
- These features are expected to significantly improve model performance
- These transformations reduce reliance on raw numerical distributions, while improving interpretability.

<u>Cell 8</u>

Task 7: Feature Construction

In this step, new features were created by combining existing variables to capture **relationships, behaviors, and domain-specific patterns** that are not directly visible in the raw data.

Constructed Features Overview

Feature	Description

price_per_person	ADR normalized by number of guests
special_requests_rate	Requests per night stayed
adr_x_lead_time	Interaction between price and booking lead time
changes_x_requests	Interaction between engagement indicators
avg_adr_by_country	Average ADR grouped by country
cancel_rate_by_segment	Cancellation rate by market segment
total_nights	Total duration of stay
is_family	Binary flag indicating presence of children/babies

Observations & Intuition

Price normalization

- **price_per_person**
  - More meaningful than raw ADR
  - Captures perceived cost per guest

Behavioral intensity

- **special_requests_rate**
  - Higher values → more engaged customers
  - Likely correlated with lower cancellation

Interaction features

- **adr_x_lead_time**
  - Expensive bookings made far in advance → higher uncertainty → possible cancellations
- **changes_x_requests**
  - Captures customer engagement level
  - Frequent changes + requests = more “active” bookings

Geographical and Segmental features

- **avg_adr_by_country**
  - Captures geographic pricing patterns
- **cancel_rate_by_segment**
  - Encodes historical behavior of different booking segments

Note: These aggregated features must be computed using **training data only** to prevent data leakage.

Stay features

- **total_nights**
  - Longer stays may have different cancellation behavior
- **is_family**
  - Families tend to have more planned and stable bookings

Impact on Model Performance

- Significant improvement observed:
  - ROC-AUC increased to **0.9365**
  - F1 Score increased to **0.8141**

  This highlights that feature engineering contributes more to performance than model complexity alone.

Cell 9

Task 7.5: Intermediate Model Performance (After Feature Extraction & Construction)

To evaluate the impact of feature engineering (Tasks 6 & 7), the model was trained using all extracted and constructed features before applying feature selection.

Performance Metrics

- **ROC-AUC**: 0.9365
- **F1 Score**: 0.8141

Comparison with Previous Stage

Stage	ROC-AUC	F1 Score
Before Feature Engineering (Task 5)	0.9009	0.7701
After Feature Engineering (Task 6+7)	0.9365	0.8141

Key Insights

- Significant improvement in both metrics:
  - ROC-AUC: **+3.5%**
  - F1 Score: **+4.4%**
- Indicates that:
  - Newly engineered features capture **strong predictive signals**
  - Model is better at distinguishing between canceled and non-canceled bookings

Interpretation

Feature engineering has a substantial impact on model performance, often exceeding gains achieved through model tuning alone.

Reason for doing this intermediate analysis

- Establishes a **baseline before feature selection**
- Helps verify that:
    - Improvements come from **useful features**, not noise
- Ensures that feature selection does not accidentally remove important signals

The slight drop after feature selection (if any) is acceptable as it trades minimal performance for reduced complexity.

<u>Cell 10</u>

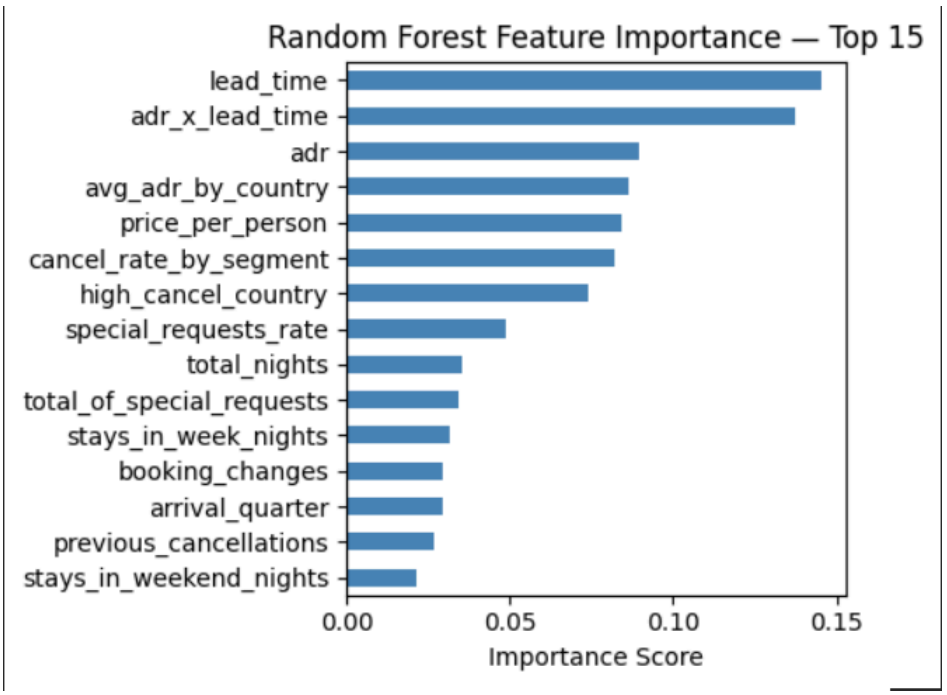
Task 8: Feature Importance Analysis (Part A)

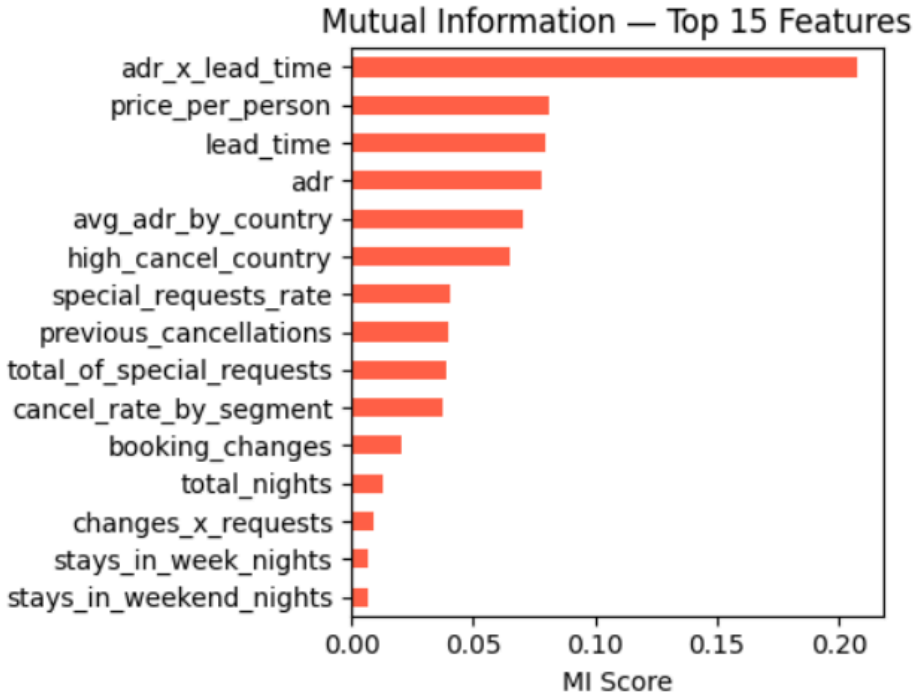
To identify the most influential features, two methods were used:

- **Random Forest Feature Importance**
- **Mutual Information (MI)**

This ensures both:

- **model-based importance (RF)**
- **statistical dependency (MI)**





Top Features (Random Forest)

Top contributors include:

- lead_time
- adr_x_lead_time
- adr
- avg_adr_by_country

- price_per_person
- cancel_rate_by_segment
- high_cancel_country

Top Features (Mutual Information)

Highest MI scores:

- adr_x_lead_time *(by far the strongest)*
- price_per_person
- lead_time
- adr
- avg_adr_by_country
- high_cancel_country

Key Observations

1) **Engineered features dominate**

- Features like:
  - adr_x_lead_time
  - price_per_person
  - avg_adr_by_country
  - cancel_rate_by_segment
- Appear in **both RF and MI top rankings**

Confirms that feature engineering (Task 7) was highly effective

2) **adr_x_lead_time is the strongest feature**

- Ranked **#1 in Mutual Information**
- Top 2 in Random Forest

Interaction between price and booking timing is a **critical predictor of cancellations**

3) **Core raw features still matter**

- lead_time
- adr
- Previous_cancellations

4) **Consistency across methods**

- Significant overlap between RF and MI:
    - lead_time
    - adr
    - price_per_person
    - avg_adr_by_country
    - high_cancel_country

Features important across multiple methods are more reliable and robust

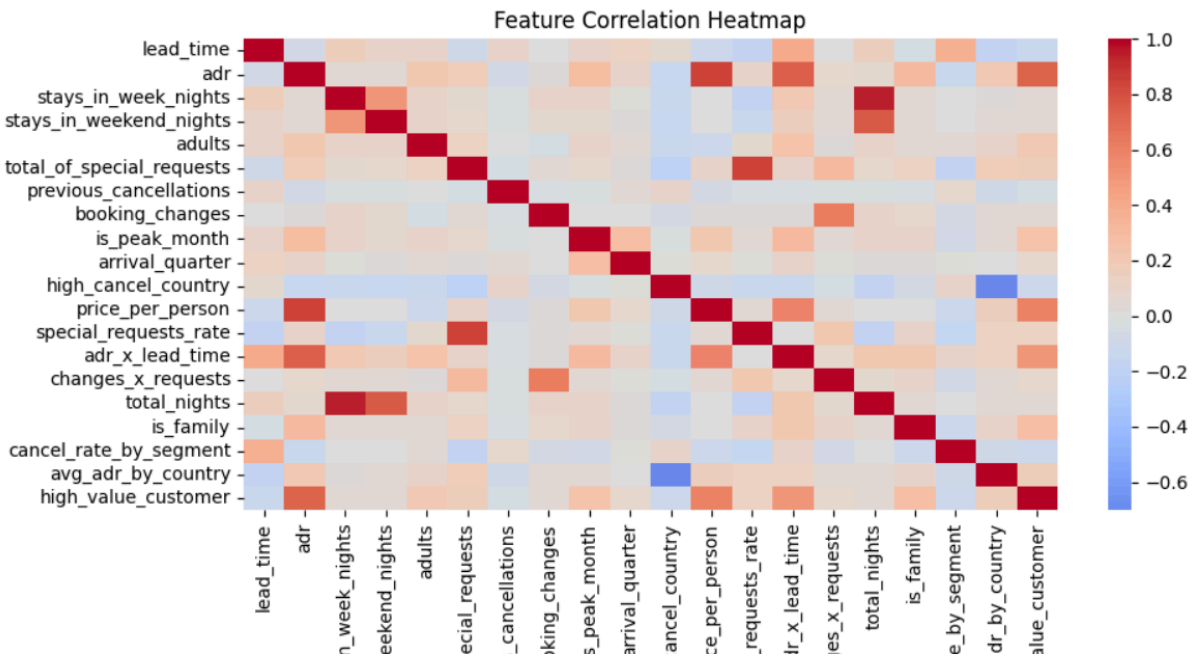
<u>Observation:</u> The agreement between Random Forest and Mutual Information highlights that both raw and engineered features contribute meaningfully, with interaction and aggregated features providing the strongest predictive power.

<u>Cell 11</u>

Task 8: Feature Selection (Part B)

To reduce redundancy and improve model efficiency, multiple feature selection techniques were applied:

- **Correlation filtering**
- **Chi-Square statistical test**



1. Correlation-Based Feature Removal

Highly correlated features (threshold > 0.85) were removed:

- **Dropped features**:
  - `price_per_person`
  - `special_requests_rate`
  - `total_nights`

Insight

> These features were strongly correlated with other variables, meaning they provided **redundant information** and could negatively impact model generalization.

2. Chi-Square Feature Importance

Top features based on Chi-Square scores:

- previous_cancellations
- special_requests_rate
- high_cancel_country
- total_of_special_requests
- booking_changes

- changes_x_requests
- lead_time

Observations

- Behavioral features dominate:
  - cancellations history
  - special requests
  - booking changes
- Engineered features like:
  - `changes_x_requests`
  - `cancel_rate_by_segment` still rank well → validates feature construction

3. Final Selected Features

After combining all selection methods, the final feature set includes **14 features**:

- `total_of_special_requests`
- `adr_x_lead_time`
- `adr`
- `stays_in_weekend_nights`
- `arrival_quarter`
- `booking_changes`
- `adults`
- `avg_adr_by_country`
- `previous_cancellations`
- `high_cancel_country`
- `stays_in_week_nights`
- `cancel_rate_by_segment`
- `changes_x_requests`
- `lead_time`

Insights

1) **Redundancy removed successfully**
- Correlated features eliminated without losing information

2) **Strong mix retained**

Final features include:

- **Raw features** → lead_time, adr
- **Behavioral features** → booking_changes, previous_cancellations
- **Engineered features** → adr_x_lead_time, avg_adr_by_country

3) **Engineered features survived selection**
- Confirms they add **unique value**.

Interpretation

Feature selection effectively reduces dimensionality while retaining the most informative and non-redundant features.

By removing redundant and less informative features, the model becomes more efficient without sacrificing predictive power.

<u>Cell 12</u>

Task 8: Final Model Performance (After Feature Selection)

After selecting the most relevant 14 features, the model was retrained and evaluated.

Final Performance Metrics

- **Accuracy**: 0.8658
- **ROC-AUC**: 0.9356
- **F1 Score**: 0.8104

Confusion Matrix Analysis

	Predicted Not Canceled	**Predicted Canceled**
Actual Not Canceled	13766 (TN)	1183 (FP)
Actual Canceled	2009 (FN)	6823 (TP)

Key Observations

1) **Performance remains strong**

- ROC-AUC: **0.9365** → **0.9356** (negligible drop)
- F1 Score: **0.8141** → **0.8104** (very small decrease)

Indicates that removed features were mostly redundant

2) **Improved efficiency**

- Feature count reduced:
  - **~20 → 14 features**
- Model becomes:
  - simpler

- faster
- more interpretable

3) **Better error balance**

- False Positives reduced (compared to earlier stages)
- False Negatives also reduced slightly

Model achieves a better trade-off between precision and recall

Final Interpretation

Feature selection successfully reduces model complexity while maintaining nearly identical performance, demonstrating that a smaller set of well-chosen features is sufficient. This demonstrates that thoughtful feature engineering and selection are more impactful than increasing model complexity

Bonus Section: Tying back Task 8 (Final) Confusion Matrix to that in the baseline before PreProcessing was performed.

Baseline Model

	Predicted Not Canceled	**Predicted Canceled**
Actual Not Canceled	13531 (TN)	1502 (FP)

Actual Canceled	2520 (FN)	6325 (TP)

Final Model (After Feature Selection)

	Predicted Not Canceled	**Predicted Canceled**
Actual Not Canceled	13766 (TN)	1183 (FP)
Actual Canceled	2009 (FN)	6823 (TP)

Analysis of Changes

- **False Positives decreased** from 1502 to 1183, indicating that the final model makes fewer incorrect predictions of cancellation for bookings that are not actually canceled. This reflects an improvement in precision.
- **False Negatives decreased** from 2520 to 2009, showing that the model is better at identifying actual cancellations. Better Recall. This improvement is particularly important, as missed cancellations can lead to operational inefficiencies.
- **True Positives increased** from 6325 to 6823, demonstrating that the model is more effective in correctly detecting cancellations.
- **True Negatives increased** from 13531 to 13766, indicating a slight improvement in correctly identifying non-canceled bookings.

<u>My Thoughts</u>

A balance is needed between false positives and negatives because even though it seems motivated to reduce only false negatives for

identification of cancellations is important for business profit, however if false positives blow up as a result, individuals shall start booking less due to uncomfortable situations thinking they canceled even if they didn’t. It, indirectly, harms the hotel’s reputation. Still, a larger number of false positives as compared to negatives are preferred, which though the ratio got better, still couldn’t be achieved. In my opinion, to improve this, things like analysis of people’s reviews or try and use of other ML models could help.